

Course - Data Engineering with Python, Azure Databricks & Power BI

What Sets Us Apart:

- Real enterprise-level projects
- Personalized Work Readiness Assessment (WRA) reports
- Two rounds of professional mock interview feedback reports
- Technical skills gap analysis with improvement roadmap
- Project evaluation reports with industry benchmarking
- Attendance and participation performance tracking
- Certification readiness assessment
- Live recorded lectures

Day wise Schedule

Day	Topic	Description
Day 1	Python Basics	Installation, IDEs, Program Flow, Keywords, Running Scripts
Day 2	File Operations & Data Structures	Indexing, Slicing, range(), Dictionaries, Sets, Files
Day 3	Functions & Advanced Python	Function Arguments, Parameters, Variable Scope, Decorators, Context Managers

Day 4	Packages & Error Handling	Package Installation, Virtual Environments, Exception Handling, Logging
Day 5	Git & Version Control	Git fundamentals, GitHub, Branching, Pull Requests, Collaboration
Day 6	Python Practice & Review	Hands-on exercises covering Days 1-5, coding challenges
Day 7	Database Handling	Python-MySQL Connectivity, CRUD Operations, SQLAlchemy
Day 8	Database Project	Building Real-World Application with MySQL
Day 9	NumPy Fundamentals	Installation, Arrays, Slicing, Multidimensional Arrays, Broadcasting
Day 10	NumPy Advanced	DTypes, Vectorized Operations, Performance Optimization
Day 11	Pandas Fundamentals	DataFrames, Series, Indexes, loc[] and iloc[], Reading Data
Day 12	DataFrame Manipulation	Accessing Values (at[], iat[]), Filtering, Sorting, Filling & Replacing
Day 13	DataFrame Operations	Merging, Joining, Concatenating, Group By Operations
Day 14	Advanced Pandas	Pivot Tables, Reshaping, Window Functions, Performance Tips

Day 15	Time Series & File I/O	DateTime Operations, Time Series Analysis, Reading/Writing Multiple Formats
Day 16	Pandas Project Day	Real-world data cleaning and analysis project
Day 17	SQL Fundamentals	Relational Databases, DDL/DML, Query Basics, Joins
Day 18	SQL Advanced	Complex Joins, Subqueries, Window Functions, CTEs
Day 19	SQL Performance	Query Optimization, Indexes, Execution Plans, Best Practices
Day 20	SQL Practice Project	Complex queries, database design, real-world scenarios
Day 21	Docker Fundamentals	Containers, Images, Dockerfile, Docker Compose, Basic Orchestration
Day 22	Apache Spark Introduction	RDDs, DataFrames, Spark Architecture, Installation & Setup
Day 23	Spark DataFrames & SQL	DataFrame API, Spark SQL, Transformations & Actions
Day 24	PySpark Advanced	Optimization, Partitioning, Broadcasting, UDFs
Day 25	Spark Practice Day	Hands-on large-scale data processing exercises

Day 26	Apache Kafka Basics	Streaming Concepts, Producers, Consumers, Topics, Partitions
Day 27	Kafka & Spark Streaming	Real-time data processing, Structured Streaming, Integration patterns
Day 28	Apache Airflow Fundamentals	DAGs, Operators, Scheduling, Dependencies, Task Management
Day 29	Airflow Advanced	Sensors, XComs, Branching, Dynamic DAGs, Best Practices
Day 30	Airflow Practice Project	Build end-to-end data pipeline with monitoring
Day 31	Azure Data Lake Storage (ADLS)	Cloud Storage Concepts, Data Organization, Access Patterns
Day 32	Azure Data Factory (ADF)	Pipeline Building, Copy Activities, Data Flows, Triggers
Day 33	Databricks Fundamentals	Workspace, Notebooks, Cluster Management, DBFS, Unity Catalog
Day 34	Databricks & PySpark	Large-scale Transformations, Databricks SQL, Performance Optimization
Day 35	Delta Lake on Databricks	Delta Lake Architecture, ACID Transactions, Time Travel, Z-Ordering
Day 36	Databricks Projects	Data Connection & Transformation Projects, Real-world Scenarios

Day 37	Databricks Advanced	Jobs, Workflows, Databricks SQL Analytics, Performance Tuning
Day 38	Cloud Data Warehouse Basics	Snowflake/BigQuery Architecture, Best Practices, Cost Optimization
Day 39	Data Quality & Monitoring	Great Expectations, Data Validation, Monitoring Pipelines, Alerts
Day 40	CI/CD for Data Pipelines	Git workflows, Testing strategies, Deployment automation
Day 41	Power BI Fundamentals	Data Connectivity, Visualizations, DAX Basics
Day 42	Power BI Advanced	Complex DAX, Performance Optimization, Dashboard Design
Day 43	End-to-End Project	Complete Pipeline: Source → ADLS → Databricks → Warehouse → BI
Day 44	Work Readiness Assessment	Final Student Interviews with Industry Experts
Day 45	Work Readiness Assessment	Final Student Interviews with Industry Experts